Low-Complexity Precoder Design for Massive MIMO LEO Satellite Multicast Communications

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***Abstract*—In this paper, we propose a low-complexity precoder design for massive multiple-input multiple-output (MIMO) low-Earth-orbit (LEO) satellite multicast communications. Both weighted sum-rate (WSR) maximization and max–min fairness (MMF) criteria are considered. By exploiting the fact that the optimal precoders lie in a low-dimensional subspace, the high-dimensional spatial-domain problem is equivalently reformulated into a reduced-dimensional form without any loss of optimality. Building upon this reformulation, efficient algorithms with guaranteed convergence are developed under the penalty dual decomposition (PDD) framework. In particular, a one-time precomputation scheme is introduced to eliminate per-iteration matrix inversions by reusing precomputed factors, thereby further reducing computational complexity. Simulation results show that the proposed design achieves superior performance with significantly lower complexity, demonstrating its effectiveness for massive MIMO LEO satellite communication systems.**

***Index Terms*—LEO satellite communications, massive MIMO, multicast transmission.**

I. Introduction

SATELLITE communication (SATCOM) has emerged as a promising solution for providing seamless coverage and ubiquitous connectivity in remote and underserved areas, thereby complementing terrestrial communication networks [1]. In particular, low-Earth-orbit (LEO) satellites, operating at altitudes between 500 and 2000 km, offer several advantages over their geostationary-orbit (GEO) counterparts, including lower propagation delay, reduced path loss, and lower deployment and operating costs [2]. Owing to these benefits, LEO satellite systems are expected to play a pivotal role in achieving truly global wireless connectivity in future networks.

Massive multiple-input multiple-output (MIMO) has been widely recognized as a key enabling technology for 5G and beyond [3]. Its application has recently been extended from terrestrial networks to SATCOM. While most existing works focus on unicast transmission [4], [5], the growing demand for content-centric services has highlighted the need for efficient multicast transmission in massive MIMO LEO systems. By simultaneously serving multiple users with the same content, multicast can significantly enhance spectral and energy efficiency [6]. Several studies have investigated multicast precoder design for LEO satellite massive MIMO communications from different perspectives. Specifically, [7] addresses the max–min fairness (MMF) criterion, while [8] considers joint precoding and user grouping to maximize the system weighted sum rate (WSR). In [9], a precoder design that maximizes energy efficiency is proposed. Although these studies provide valuable insights, most existing precoder designs have prohibitive computational complexity, which hinders their scalability to massive MIMO LEO satellite communication systems.

In this paper, we propose a low-complexity precoder design for massive MIMO LEO satellite multicast communications. By exploiting the fact that the optimal precoders lie in a low-dimensional subspace, the high-dimensional spatial-domain problem is equivalently reformulated into a reduced-dimensional form without any loss of optimality. Building upon this reformulation, efficient algorithms with guaranteed convergence are developed under the PDD framework [10]. In particular, a one-time precomputation scheme is introduced to eliminate per-iteration matrix inversions by reusing precomputed factors, thereby further reducing computational complexity. Simulation results show that the proposed design achieves superior performance with significantly lower complexity, demonstrating its effectiveness for massive MIMO LEO satellite communication systems.

II. System Model

We consider a full-frequency-reuse (FFR) downlink LEO satellite system supporting multigroup multicast transmission. A single LEO satellite at altitude H serves G groups, whose indices are denoted as $\mathcal{G} \triangleq \{1, \ldots, G\}$. The set of all UTs is denoted as $\mathcal{K} \triangleq \{1, \ldots, K\}$, and the subset of UTs belonging to group g is denoted by $\mathcal{K}_g \subseteq \mathcal{K}$ with $|\mathcal{K}_g| = K_g$. Each UT belongs to exactly one group, and $I(k)$ denotes the group index of user k. All UTs in the same group share the same multicast data stream. The satellite is equipped with a uniform

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planar array (UPA), while each UT has a single antenna. The UPA comprises $M = M_x \times M_y$ antennas, where M_x and M_y are the numbers of antennas along the x- and y-axes, respectively.

A. Channel Model

The channel response vector of the k-th UT at time t and frequency f is given by [4]

$$\mathbf{h}_k(t,f) = \tilde{h}_k(t,f)\,\mathbf{g}_k, \tag{1}$$

where $\mathbf{g}_k \in \mathbb{C}^{M\times 1}$ denotes the UPA array response vector of the k-th UT, expressed as $\mathbf{g}_k = \mathbf{g}_x(\theta_k^x) \otimes \mathbf{g}_y(\theta_k^y)$, with $\theta_k^x = \sin\phi_k^y \cos\phi_k^x$ and $\theta_k^y = \cos\phi_k^y$, where ϕ_k^x and ϕ_k^y denote the azimuth and elevation angles of the k-th UT, respectively. The steering vector on dimension $d \in \{x, y\}$ is given by

$$\mathbf{g}_d(\theta_k^d) = \frac{1}{\sqrt{M_d}}\left[1,\, e^{-j\pi\theta_k^d},\, \ldots,\, e^{-j\pi(M_d-1)\theta_k^d}\right]^T \in \mathbb{C}^{M_d\times 1}, \tag{2}$$

After compensating for Doppler shifts due to satellite motion and for the minimum propagation delay at the receiver [4], $\tilde{h}_k(t,f)$ is modeled as

$$\tilde{h}_k(t,f) = \sum_{p=1}^{P_k} \alpha_{k,p}\, e^{j2\pi(t\nu_{k,p} - f\tau_{k,p})}, \tag{3}$$

where P_k is the number of propagation paths, and $\alpha_{k,p}$, $\nu_{k,p}$, and $\tau_{k,p}$ denote the complex path gain, Doppler shift, and delay of the p-th path, respectively. Here, $\tilde{h}_k(t,f)$ is obtained from the time-varying impulse response $h_k(t,\tau)$ by applying the Fourier transform with respect to the delay variable τ, i.e., $\tilde{h}_k(t,f) = \int h_k(t,\tau)\, e^{-j2\pi f\tau} d\tau$, where $h_k(t,\tau) \triangleq \sum_{p=1}^{P_k} \alpha_{k,p} e^{j2\pi\nu_{k,p}t}\delta(\tau - \tau_{k,p})$.

At any given time and frequency, $\tilde{h}_k(t,f)$ follows a Rician distribution with factor κ_k, and its average power satisfies $\mathbb{E}\{|\tilde{h}_k(t,f)|^2\} = \beta_k$. Following [4], we refer to β_k and $\mathbf{g}_k$ as the statistical channel state information (CSI) of UT k.

B. Signal Model

Suppose that orthogonal frequency division multiplexing (OFDM) is employed to support wideband transmission in the LEO SATCOM system. The received signal of the k-th UT on subcarrier r of OFDM symbol s is given by

$$y_{k,s,r} = \mathbf{h}_{k,s,r}^H \mathbf{w}_{I(k),s,r} s_{I(k),s,r} + \sum_{g\neq I(k)} \mathbf{h}_{k,s,r}^H \mathbf{w}_{g,s,r} s_{g,s,r} + z_{k,s,r}, \tag{4}$$

where $\mathbf{h}_{k,s,r} \triangleq \mathbf{h}_k(sT_{\mathrm{sym}}, r\Delta_f)$, and T_{sym} and Δ_f denote the OFDM symbol duration and subcarrier spacing, respectively. Here, all UTs belonging to the same multicast group g share a common data symbol $s_{g,s,r}$ and a common precoding vector $\mathbf{w}_{g,s,r}$. The noise term is modeled as $z_{k,s,r} \sim \mathcal{CN}(0, \sigma_{k,s,r}^2)$. For simplicity, the indices s and r are omitted hereafter when no ambiguity arises.

C. Problem Formulation

Owing to the high mobility of the LEO satellite, the transmitter is assumed to have only statistical CSI, characterized by β_k and $\mathbf{g}_k$, in this paper. Accordingly, for multigroup multicast communications, the ergodic rate of group g is defined as in [7].

$$R_g = \min_{k\in\mathcal{K}_g} R_k(\{\mathbf{w}_g\}_{g=1}^G), \tag{5}$$

where the user rate is given by

$$R_k(\{\mathbf{w}_g\}_{g=1}^G) \triangleq \mathbb{E}\left\{\log\left(1 + \frac{|\mathbf{h}_k^H \mathbf{w}_{I(k)}|^2}{\sum_{g\neq I(k)} |\mathbf{h}_k^H \mathbf{w}_g|^2 + \sigma_k^2}\right)\right\}. \tag{6}$$

Since the ergodic rate in (5) generally lacks a closed-form expression, it is difficult to optimize directly. Following [4], we approximate R_g with its upper bound $\overline{R}_g^{\mathrm{ub}} \triangleq \min_{k\in\mathcal{K}_g} \overline{R}_k^{\mathrm{ub}}(\{\mathbf{w}_g\}_{g=1}^G)$, where

$$\overline{R}_k^{\mathrm{ub}}(\{\mathbf{w}_g\}_{g=1}^G) \triangleq \log\left(1 + \frac{|\mathbf{g}_k^H \mathbf{w}_{I(k)}|^2 \beta_k}{\sum_{g\neq I(k)} |\mathbf{g}_k^H \mathbf{w}_g|^2 \beta_k + \sigma_k^2}\right). \tag{7}$$

This upper bound has been verified to be tight in [7].

In this work, we consider the precoder design under both weighted sum-rate (WSR) and max–min fairness (MMF) criteria. The precoder design problem under the WSR criterion is formulated as

$$\mathcal{P}_1: \quad \max_{\{\mathbf{w}_g\}_{g=1}^G} \sum_{g\in\mathcal{G}} w_g \overline{R}_g^{\mathrm{ub}}(\{\mathbf{w}_g\}_{g=1}^G) \tag{8a}$$

$$\text{s.t.} \quad \sum_{g\in\mathcal{G}} \mathbf{w}_g^H \mathbf{w}_g \le P_T. \tag{8b}$$

The precoder design problem under the MMF criterion is formulated as

$$\mathcal{P}_2: \quad \max_{\{\mathbf{w}_g\}_{g=1}^G} \min_{g\in\mathcal{G}} \overline{R}_g^{\mathrm{ub}}(\{\mathbf{w}_g\}_{g=1}^G) \tag{9a}$$

$$\text{s.t.} \quad \text{(8b)}. \tag{9b}$$

Both $\mathcal{P}_1$ and $\mathcal{P}_2$ are non-convex and NP-hard, and they are more difficult than the unicast scenario. Moreover, the precoder dimension scales with M, making the optimization computationally prohibitive in massive MIMO LEO satellite communication systems. This motivates the development of low-complexity precoder design.

III. Method

A. Precoder Design under the WSR Criterion

In this subsection, we elaborate on the precoder design under the WSR criterion.

Theorem 1: *Let* $\mathbf{G} \triangleq [\mathbf{g}_1, \cdots, \mathbf{g}_K] \in \mathbb{C}^{M\times K}$. *The optimal solution to* $\mathcal{P}_1$ *lies in the column space of* $\mathbf{G}$.

The result above is consistent with [11]. Consequently, without loss of optimality, we can restrict the spatial-domain precoder to

$$\mathbf{w}_g = \mathbf{G}\mathbf{u}_g, \quad \forall g \in \mathcal{G}. \tag{10}$$

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Note that, without loss of generality, we assume that the steering vectors associated with different users are linearly independent, since otherwise a maximal linearly independent subset can always be selected.

By substituting (10) into (8) and introducing the auxiliary variables $\{r_g\}_{g=1}^G$, $\mathcal{P}_1$ can be equivalently reformulated as

$$\max_{\{\mathbf{u}_g\}_{g=1}^G} \quad \sum_{g\in\mathcal{G}} w_g r_g \tag{11a}$$

$$\text{s.t.} \quad \overline{R}_k^{\text{ub}}(\{\mathbf{u}_g\}_{g=1}^G) \geq r_{I(k)}, \quad \forall k \in \mathcal{K}, \tag{11b}$$

$$\sum_{g\in\mathcal{G}} \mathbf{u}_g^H \mathbf{B} \mathbf{u}_g \leq P_T, \tag{11c}$$

where $\mathbf{B} \triangleq \mathbf{G}^H\mathbf{G} \in \mathbb{C}^{K\times K}$ is Hermitian positive definite, since the column vectors of $\mathbf{G}$ are linearly independent, and $\overline{R}_k^{\text{ub}}(\{\mathbf{u}_g\}_{g=1}^G) \triangleq \log\left(1+\frac{|\mathbf{a}_k^H\mathbf{u}_{I(k)}|^2\beta_k}{\sum_{g'\neq I(k)}|\mathbf{a}_k^H\mathbf{u}_{g'}|^2\beta_k+\sigma_k^2}\right)$ with $\mathbf{a}_k \triangleq \mathbf{G}^H\mathbf{g}_k \in \mathbb{C}^K$.

Despite the dimension is reduced, the problem in (11) is still hard to solve. To solve it effectively, we adopt the penalty dual decomposition (PDD) method [10]. We first introduce the following auxiliary variables

$$p_{k,g} = \mathbf{a}_k^H\mathbf{u}_g, \quad \forall k\in\mathcal{K}, \forall g\in\mathcal{G}, \tag{12a}$$

$$\bar{r}_k = r_{I(k)}, \quad \forall k\in\mathcal{K}, \tag{12b}$$

and then the constraints in (11b) become

$$\bar{r}_k - \overline{R}_k^{\text{ub}}(\mathbf{p}_k) \leq 0, \quad \forall k\in\mathcal{K}, \tag{13a}$$

where $\mathbf{p}_k \triangleq [p_{k,1},\ldots,p_{k,G}]^T \in \mathbb{C}^G$. Therefore, the augmented lagrangian (AL) problem corresponding to (11) is formulated as

$$\mathcal{P}_3: \quad \max_{\{\mathbf{U},\mathbf{P},\mathbf{Z},\mathbf{v},\mathbf{r},\bar{\mathbf{r}}\}} \quad \mathcal{L}(\mathbf{U},\mathbf{P},\mathbf{Z},\mathbf{v},\mathbf{r},\bar{\mathbf{r}}) \tag{14a}$$

$$\text{s.t.} \quad (11c),(12a),(12b),(13a), \tag{14b}$$

where $\mathbf{U} \triangleq [\mathbf{u}_1,\ldots,\mathbf{u}_G] \in \mathbb{C}^{K\times G}$, $\mathbf{P} \triangleq [\mathbf{p}_1,\ldots,\mathbf{p}_K] \in \mathbb{C}^{G\times K}$, $\bar{\mathbf{r}} \triangleq [\bar{r}_1,\ldots,\bar{r}_K] \in \mathbb{C}^{K\times 1}$, $\mathbf{r} \triangleq [r_1,\cdots,r_g] \in \mathbb{R}^{G\times 1}$, and the AL function is given by

$$\mathcal{L}_\rho(\mathbf{U},\mathbf{P},\mathbf{r},\bar{\mathbf{r}},\mathbf{Z},\mathbf{v}) \triangleq \sum_{g\in\mathcal{G}} w_g r_g - \frac{1}{2\rho}\sum_{k\in\mathcal{K}} |\bar{r}_k - r_{I(k)} + \rho v_k|^2 - \frac{1}{2\rho}\sum_{k\in\mathcal{K}}\sum_{g\in\mathcal{G}} |p_{k,g} - \mathbf{a}_k^H\mathbf{u}_g + \rho z_{k,g}|^2, \tag{15}$$

where $\mathbf{v} \triangleq [v_1,\ldots,v_K] \in \mathbb{R}^K$ and $\mathbf{Z} \triangleq [z_{k,g}] \in \mathbb{C}^{G\times K}$ denote the dual variables, and $\rho > 0$ is the penalty parameter.

In $\mathcal{P}_3$, the variables can be grouped into four blocks, namely $\{\mathbf{P},\bar{\mathbf{r}}\}$, $\mathbf{r}$, $\mathbf{U}$, and $\{\mathbf{v},\mathbf{Z}\}$, which are decoupled in the constraints. Exploiting this block structure, $\mathcal{P}_3$ can be efficiently addressed using the BSUM approach [12]. Let $\rho^{(t)}$ and $\mathbf{v}^{(t)},\mathbf{Z}^{(t)}$ denote the penalty parameter and dual variables at the t-th outer iteration, respectively, and let $\mathbf{U}^{(t,n)}$, $\mathbf{P}^{(t,n)}$, $\bar{\mathbf{r}}^{(t,n)}$, and $\mathbf{r}^{(t,n)}$ denote the corresponding primal variables at the n-th inner iteration within the t-th outer loop, respectively.

1) Update of $\mathbf{U}$*:* The subproblem for updating $\mathbf{U}$ is formulated as

$$\min_{\{\mathbf{u}_g\}_{g=1}^G} \quad \sum_{g\in\mathcal{G}}\sum_{k\in\mathcal{K}} |p_{k,g}^{(t,n)} - \mathbf{a}_k^H\mathbf{u}_g + \rho^{(t)} z_{k,g}^{(t)}|^2 \tag{16a}$$

$$\text{s.t.} \quad (11c). \tag{16b}$$

This subproblem is convex, and its optimal solution can be derived by the KKT conditions [13], as given by

$$\mathbf{u}_g^* = \begin{cases} \tilde{\mathbf{u}}_g(0), & \bar{P} \leq P_T, \\ \tilde{\mathbf{u}}_g(\lambda^*), & \bar{P} > P_T, \end{cases} \tag{17}$$

where $\tilde{\mathbf{u}}_g(\lambda^*) \triangleq (\mathbf{A}+\lambda^*\mathbf{B})^{-1}\bar{\mathbf{a}}_g^{(t,n)}$, $\tilde{\mathbf{u}}_g(0) \triangleq \mathbf{A}^\dagger\bar{\mathbf{a}}_g^{(t,n)}$, $\bar{P} \triangleq \sum_{g\in\mathcal{G}} \tilde{\mathbf{u}}_g^H(0)\mathbf{B}\tilde{\mathbf{u}}_g(0)$. Here, $\bar{\mathbf{a}}_g^{(t,n)} \triangleq \sum_{k\in\mathcal{K}} \mathbf{a}_k q_{k,g}^{(t,n)}$, $q_{k,g}^{(t,n)} \triangleq p_{k,g}^{(t,n)} + \rho^{(t)} z_{k,g}^{(t)}$, $\mathbf{A} \triangleq \sum_{k\in\mathcal{K}} \mathbf{a}_k\mathbf{a}_k^H$, and λ^* denotes the positive root of equation $\sum_{g\in\mathcal{G}} \tilde{\mathbf{u}}_g(\lambda)^H\mathbf{B}\tilde{\mathbf{u}}_g(\lambda) = P_T$.

Notably, computing $\tilde{\mathbf{u}}_g^*$ involves large matrix inversion. To amortize this cost, we herein propose a precomputation scheme. Specifically, for the computation of $\tilde{\mathbf{u}}_g(0)$, we first precompute $\mathbf{A}^\dagger$ with a one-time cost of $\mathcal{O}(K^3)$. Then, each computation of $\tilde{\mathbf{u}}_g(0)$ is reduced to $\mathcal{O}(K^2)$. For the computation of $\tilde{\mathbf{u}}_g(\lambda^*)$, we propose a congruence simultaneous diagonalization [14] scheme.

Lemma 1: Let $\mathbf{X} \succeq \mathbf{0} \in \mathbb{C}^{N\times N}$ and $\mathbf{Y} \succ \mathbf{0} \in \mathbb{C}^{N\times N}$. There exist a nonsingular $\mathbf{Z} \in \mathbb{C}^{N\times N}$ and a diagonal $\mathbf{D} \in \mathbb{C}^{N\times N}$ such that

$$\mathbf{Z}^{-H}\mathbf{X}\mathbf{Z}^{-1} = \mathbf{D}, \quad \mathbf{Z}^{-H}\mathbf{Y}\mathbf{Z}^{-1} = \mathbf{I}_N. \tag{18}$$

where $\mathbf{I}_N$ is the $N\times N$ identity matrix.

Applying Lemma 1 to $\mathbf{X} = \mathbf{A} \succeq \mathbf{0} \in \mathbb{C}^{K\times K}$ and $\mathbf{Y} = \mathbf{B} \succ \mathbf{0} \in \mathbb{C}^{K\times K}$ yields

$$\tilde{\mathbf{u}}_g(\lambda) = (\mathbf{A}+\lambda\mathbf{B})^{-1}\bar{\mathbf{a}}_g^{(t,n)} \tag{19a}$$

$$= \mathbf{Z}(\mathbf{D}_A + \lambda\mathbf{I}_K)^{-1}\mathbf{Z}^H\bar{\mathbf{a}}_g^{(t,n)}, \tag{19b}$$

where $\mathbf{Z} \in \mathbb{C}^{K\times K}$ is nonsingular and $\mathbf{D}_A$ is diagonal. Note that $\mathbf{Z}$ and $\mathbf{D}_A$ can be precomputed once and reused throughout all iterations. To obtain $\mathbf{Z}$ and $\mathbf{D}_A$, we first perform the Cholesky factorization $\mathbf{B} = \mathbf{L}\mathbf{L}^H$, followed by the eigendecomposition $\mathbf{S} = \mathbf{L}^{-1}\mathbf{A}\mathbf{L}^{-H} = \mathbf{Q}\mathbf{D}_A\mathbf{Q}^H$, and finally set $\mathbf{Z} = \mathbf{L}^{-H}\mathbf{Q}$. This preprocessing requires a one-time cost of $\mathcal{O}(K^3)$, after which evaluating (19) involves only diagonal scaling and two matrix–vector multiplications, resulting in an $\mathcal{O}(K^2)$ cost.

2) Update of $\{\mathbf{P},\bar{\mathbf{r}}\}$*:* The subproblem for the update of $\{\mathbf{P},\bar{\mathbf{r}}\}$ can be divided into K subproblems for each UT, as given by

$$\begin{aligned} \min_{\mathbf{p}_k,\bar{r}_k} \quad & \sum_{g\in\mathcal{G}} |p_{k,g} - d_{k,g}^{(t,n+1)}|^2 + |\bar{r}_k - e_k^{(t,n)}|^2 \\ \text{s.t.} \quad & \bar{r}_k - \overline{R}_k^{\text{ub}}(\mathbf{p}_k) \leq 0, \end{aligned} \tag{20}$$

where $d_{k,g}^{(t,n+1)} \triangleq \mathbf{a}_k^H\mathbf{u}_g^{(t,n+1)} - \rho^{(t)}z_{k,g}^{(t)}$, $e_k^{(t,n)} \triangleq r_{I(k)}^{(t,n)} - \rho^{(t)}v_k^{(t)}$. This problem is nonconvex due to the nonconvex constraint, making it difficult to solve directly. To address this challenge, we propose to obtain a locally optimal solution by

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iteratively solving a sequence of convex subproblems, where $\overline{R}_k^{\rm ub}(\mathbf{p}_k)$ is replaced with a tight convex lower bound.

Lemma 2: *For any feasible point $\hat{\mathbf{p}}_k \triangleq [\hat{p}_{k,1},\ldots,\hat{p}_{k,G}]^T$, $\overline{R}_k^{\rm ub}(\mathbf{p}_k)$ is globally lower-bounded by the concave quadratic function $\underline{R}_k(\mathbf{p}_k,\hat{\mathbf{p}}_k)$, such that*

$$\begin{aligned}\overline{R}_k^{\rm ub}(\mathbf{p}_k) &\geq \underline{R}_k(\mathbf{p}_k,\hat{\mathbf{p}}_k)\\ &\triangleq \alpha_k + 2{\rm Re}\big\{(\mathbf{b}_k)^H\mathbf{p}_k\big\} - \mathbf{p}_k^H\mathbf{C}_k\mathbf{p}_k,\end{aligned} \tag{21}$$

where $a_k\in\mathbb{R}$, $\mathbf{b}_k\in\mathbb{C}^G$, and $\mathbf{C}_k\in\mathbb{C}^{G\times G}$ are given by

$$\alpha_k \triangleq \overline{R}_k^{\rm ub}(\hat{\mathbf{p}}_k) - \frac{|\hat{p}_{k,I(k)}|^2}{\xi_k} - \frac{2|\hat{p}_{k,I(k)}|^2\sigma_k^2}{\beta_k\zeta_k\xi_k}, \tag{22}$$

$$[\mathbf{b}_k]_i \triangleq \begin{cases}\dfrac{\hat{p}_{k,I(k)}}{\xi_k}, & i=I(k),\\ 0, & i\neq I(k),\end{cases} \tag{23}$$

and $\mathbf{C}_k = c_k\mathbf{I}_G$, with $c_k \triangleq \dfrac{|\hat{p}_{k,I(k)}|^2}{\zeta_k\xi_k}$, $\xi_k \triangleq \sum_{g'\neq I(k)}|\hat{p}_{k,g'}|^2+\sigma_k^2/\beta_k$, $\zeta_k \triangleq \xi_k+|\hat{p}_{k,I(k)}|^2$. Moreover, we have $\dfrac{\partial\overline{R}_k^{\rm ub}(\mathbf{p},\hat{\mathbf{p}}_k)}{\partial\mathbf{p}}\Big|_{\mathbf{p}=\hat{\mathbf{p}}} = \dfrac{\partial\underline{R}_k(\mathbf{p},\hat{\mathbf{p}}_k)}{\partial\mathbf{p}}\Big|_{\mathbf{p}=\hat{\mathbf{p}}}$.

Let $\mathbf{p}_k^{(m)}$ be the optimal solution obtained in the m-th iteration, by replacing $\overline{R}_k^{\rm ub}(\mathbf{p}_k)$ with $\underline{R}_k(\mathbf{p}_k,\mathbf{p}_k^{(m)})$, the $m+1$-th subproblem is formulated as

$$\min_{\mathbf{p}_k,\bar{r}_k} \quad \sum_{g\in\mathcal{G}}|p_{k,g}-d_{k,g}^{(t,n+1)}|^2+|\bar{r}_k-e_k^{(t,n)}|^2 \tag{24a}$$

$$\text{s.t.} \quad \bar{r}_k - \underline{R}_k(\mathbf{p}_k,\mathbf{p}_k^{(m)}) \leq 0. \tag{24b}$$

The problem above is convex, and its solution can be derived from the KKT conditions [13], as given by

$$r_k^* = r_{I(k)}^{(t,n)} - \rho^{(t)}v_k^{(t)} - \frac{\mu^*}{2}, \tag{25a}$$

$$\mathbf{p}_k^* = (\mathbf{I}+\mu^*\mathbf{C}_k^{(m)})^{-1}(\mathbf{d}_k^{(t,n+1)}+\mu^*\mathbf{b}_k^{(m)}), \tag{25b}$$

where $\mathbf{b}_k^{(m)},\mathbf{C}_k^{(m)}$ are defined in Lemma 2 by replacing $\hat{\mathbf{p}}_k$ with $\mathbf{p}_k^{(m)}$, $\mathbf{d}_k^{(t,n+1)} \triangleq [d_{k,1}^{(t,n+1)},\ldots,d_{k,G}^{(t,n+1)}]^T$, μ^* is 0 when the constraint is inactive (i.e., $r_{I(k)}^{(t,n)}-\rho^{(t)}v_k^{(t)}-\underline{R}_k(\mathbf{d}_k^{(t,n+1)})\leq 0$) and the unique positive root of the secular equation $\phi(\mu)=0$ with $\phi(\mu)\triangleq\left(r_{I(k)}^{(t,n)}-\rho^{(t)}v_k^{(t)}-\frac{\mu}{2}\right)-\underline{R}_k\left((\mathbf{I}+\mu\mathbf{C}_k^{(m)})^{-1}(\mathbf{d}_k+\mu\mathbf{b}_k^{(m)})\right)$ when the constraint is active.

3) Update of r: The subproblem for updating r is formulated as

$$\max_{\mathbf{r}} \quad \sum_{g\in\mathcal{G}}w_gr_g - \frac{1}{2\rho^{(t)}}\sum_{k\in\mathcal{K}}|\bar{r}_k^{(t,n+1)}-r_{I(k)}+\rho^{(t)}v_k^{(t)}|^2. \tag{26a}$$

This subproblem is convex, and its optimal solution is given by

$$r_g^* = (w_g\rho^{(t)}+\sum_{k\in\mathcal{K}_g}(\bar{r}_k^{(t,n+1)}+\rho^{(t)}v_k^{(t)}))/|\mathcal{K}_g|. \tag{27}$$

Algorithm 1 Algorithm for Solving $\mathcal{P}_1$.

```
Initialize t = 0, U^(t), P^(t), r^(t), r̄^(t), Z^(t), v^(t), ρ^(t).
repeat
    Initialize n = 0, W^(t,n) = W^(t), P^(t,n) = P^(t).
    repeat
        Update U^(t,n+1) using (17),
        Update P^(t,n+1) and r̄^(t,n+1) using Algorithm 2,
        Update r^(t,n+1) using (27),
        n = n + 1,
    until Δ_L^(t,n) ≤ ε_1 or n ≥ N_1^max.
    U^(t+1) = U^(n), P^(t+1) = P^(n), r̄^(t+1) = r̄^(n), r^(t+1) = r^(n),
    if Δ_L^(t+1) ≤ η then
        Update Z^(t+1) using (28a), update v^(t+1) using (28b),
        ρ^(t+1) = ρ^(t),
    else
        Z^(t+1) = Z^(t), v^(t+1) = v^(t), ρ^(t+1) = ξρ^(t).
    end if
    t = t + 1,
until Λ^(t) ≤ ε_2 or t ≥ N_2^max.
return U^(t), r^(t).
```

4) Update of the Dual Variables $\mathbf{Z}$ and $\mathbf{v}$: The update of the dual variables $\mathbf{Z},\mathbf{v}$ are given by

$$z_{k,g}^{(t+1)} = z_{k,g}^{(t)} + \frac{1}{\rho^{(t)}}\left(p_{k,g}^{(t+1)} - \mathbf{a}_k^H\mathbf{u}_g^{(t+1)}\right), \tag{28a}$$

$$v_k^{(t+1)} = v_k^{(t)} + \frac{1}{\rho^{(t)}}\left(\bar{r}_k^{(t+1)} - r_{I(k)}^{(t+1)}\right). \tag{28b}$$

5) Overall Solution: The overall algorithm is summarized in Algorithm 1, where $\Delta_L^{(t,n)} \triangleq |\mathcal{L}_\rho^{(t,n)}-\mathcal{L}_\rho^{(t,n-1)}|/|\mathcal{L}_\rho^{(t,n-1)}|$, $\mathcal{L}_\rho^{(t,n)} \triangleq \mathcal{L}_\rho(\mathbf{U}^{(t,n)},\mathbf{P}^{(t,n)},\mathbf{r}^{(t,n)},\bar{\mathbf{r}}^{(t,n)},\mathbf{Z}^{(t)},\mathbf{v}^{(t)})$ and

$$\Lambda^{(t)} \triangleq \max_{k\in\mathcal{K},g\in\mathcal{G}}\{|\bar{r}_k^{(t)}-r_{I(k)}^{(t)}|,|p_{k,g}^{(t)}-\mathbf{a}_k^H\mathbf{u}_g^{(t)}|\}. \tag{29}$$

The $\Delta_{PR}^{(m)}$ in Algorithm 2 is defined as

$$\Delta_{PR}^{(m)} \triangleq \frac{\left|F\left(\mathbf{p}_k^{(m)},\bar{\mathbf{r}}_k^{(m)}\right)-F\left(\mathbf{p}_k^{(m-1)},\bar{\mathbf{r}}_k^{(m-1)}\right)\right|}{\left|F\left(\mathbf{p}_k^{(m-1)},\bar{r}_k^{(m-1)}\right)\right|}, \tag{30}$$

where F denotes the objective function in (20).

Theorem 2: *The solution sequence generated by Algorithm 1 is bounded and hence admits at least one limit point; moreover, any limit point is a KKT point of $\mathcal{P}_1$.*

Proof Sketch: In each outer iteration, we solve the AL subproblem via BSUM to a KKT point. Following [10], the PDD convergence conditions hold; thus any accumulation point is a KKT solution of $\mathcal{P}_1$. ■

B. Precoder Design under the MMF Criterion

In this subsection, we consider the precoder design under the MMF criterion. Since $\mathcal{P}_2$ can be addressed in a manner similar to the WSR case, we only outline the key steps.

Following Theorem 1, an optimal solution to $\mathcal{P}_2$ lies in the column space of $\mathbf{G}$. Hence, the precoder can be parameterized

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Algorithm 2 Algorithm for the Update of $\{\mathbf{P}, \bar{\mathbf{r}}\}$.

1: Initialize $m \leftarrow 0$, $\mathbf{P}^{(m)} \leftarrow \mathbf{P}^{(t,n)}$, $\bar{\mathbf{r}}^{(m)} \leftarrow \bar{\mathbf{r}}^{(t,n)}$
2: **for all** UT k **do**
3: **repeat**
4: Update $\mathbf{p}_k^{(m+1)}$ and $\bar{r}_k^{(m+1)}$ according to (25)
5: $m \leftarrow m+1$
6: **until** $\Delta_{PR}^{(m)} \leq \epsilon_3$ **or** $m \geq N_3^{\max}$
7: **end for**
8: **return** $\mathbf{P}^{(m)}$ and $\bar{\mathbf{r}}^{(m)}$

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Number of UEs k	50
Number of groups	5
Radius of the earth R_E	6378 km
Orbit altitude of LEO satellites H	1000 km
Number of subcarriers N_{sc}	1024
Carrier frequency f_c	4 GHz
Noise temperature T_n	290 K
Bandwidth B	50 MHz
Antenna spacing	$\frac{\lambda}{2}$
Maximum nadir angle $\theta_{\max}$	45°
Per-antenna gain $G_{\text{sat}}, G_{\text{ut}}$	6 dBi, 0 dBi
Total transmit power P_T	5 dBW $\sim$ 25 dBW

as $\mathbf{w}_g = \mathbf{G}\mathbf{u}_g, \forall g \in \mathcal{G}$, and optimization is performed over $\{\mathbf{u}_g\}_{g=1}^G$. Introducing auxiliary variables as in (13a) and setting $\bar{r}_k = r$ for all k, $\mathcal{P}_2$ can be equivalently expressed in the AL form

$$\max_{\mathbf{U},\mathbf{P},r,\bar{\mathbf{r}},\mathbf{Z},\mathbf{v}} \quad \mathcal{L}_\rho(\mathbf{U}, \mathbf{P}, r, \bar{\mathbf{r}}, \mathbf{Z}, \mathbf{v}) \tag{31}$$

where $\mathcal{L}_\rho(\cdot)$ is defined analogously to (15), and the associated constraints are similar to those in (14) and thus omitted for brevity.

The updates of $\mathbf{U}$ and $\{\mathbf{P}, \bar{\mathbf{r}}\}$ follow the same structures as those in the WSR case, with $\mathbf{U}$ given by (17) and $\{\mathbf{P}, \bar{\mathbf{r}}\}$ obtained by solving a relaxed quadratic subproblem similar to (24). The scalar variable r is updated in closed form as

$$r^* = \frac{\rho^{(t)} + \mathbf{1}_K^T(\bar{\mathbf{r}}^{(t,n+1)} + \rho^{(t)}\mathbf{v}^{(t)})}{K}. \tag{32}$$

The overall algorithm follows the same structure as Algorithm 1, except for the updates of $\bar{\mathbf{r}}$ and r, and is thus omitted for brevity.

C. Complexity Analysis

The complexities of the proposed precoder design scheme under both WSR and MMF criteria are $\mathcal{O}\left(K^3 + I_{\text{outer}} I_{\text{inner}} \left(GK^2 + I_P GK\right)\right)$, where $I_{\text{outer}}, I_{\text{inner}}$ are the numbers of iterations of the outer and the inner loops in Algorithm 1, respectively, and I_P is the number of iterations in Algorithm 2.

IV. SIMULATION RESULTS

In this section, we evaluate the proposed precoder design through simulations. The main simulation parameters are listed in Table I. The UTs are generated in the coverage area of the satellite with uniform distribution. The groups are assumed to be of equal weights for simplicity. The satellite orbit radius is $R_s = H + R_e$, where H is the orbit altitude of the satellite and R_e is the earth radius. The elevation angle is calculated as $\alpha_k = \cos^{-1}\left(\frac{R_s \sin(\vartheta_k)}{R_e}\right)$, where ϑ_k is the nadir angle of UT k. The distance between UT k and the satellite is $D_k = -\sin(\alpha_k)R_e + \sqrt{2R_eH + \left(\sin(\alpha_k)R_e\right)^2 + H^2}$. Let G_{ut} denote the antenna gain at UTs, and G_{sat} denote the per-antenna gain at satellite. For different elevation angles, path loss and shadow fading are modeled according to those as in [4].

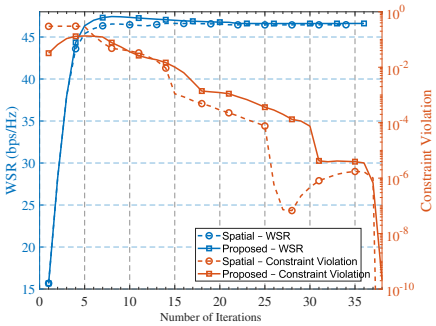


Fig. 1. Convergence performance of the proposed precoder design.

Algorithmic hyperparameters are fixed across experiments as $\rho^{(0)} = 0.4$, $\eta = \epsilon_1 = 10^{-4}$, $\xi = 0.95$, $\epsilon_2 = 10^{-8}$, $\epsilon_3 = 5 \times 10^{-5}$, $N_1^{\max} = 20$, and $N_2^{\max} = 100$, $N_3^{\max} = 10$. During performance evaluation, the group rate is evaluated using (5) rather than the upper-bound approximation. All simulations are conducted on a workstation equipped with an Intel(R) Xeon(R) Gold 6226R CPU

For comparisons, we consider following baselines:

- **Scheme 1 (spatial-domain)** optimizes the M-dimensional beamformers $\{\mathbf{w}_g\}_{g=1}^G$ under the same objective and power constraint, without exploiting the low-dimensional structure, resulting in complexity $\mathcal{O}\big(I_{\text{outer}} I_{\text{inner}}(GM^3 + I_P GM)\big)$.
- **Scheme 2 (spatial-domain + precomputation)** follows Scheme 1 but incorporates the proposed precomputation technique, which lowers the cost to $\mathcal{O}\big(M^3 + I_{\text{outer}} I_{\text{inner}}(GM^2 + I_P GM)\big)$.
- **Scheme 3 (proposed without precomputation)** adopts the low-dimensional reformulation without precomputation, with complexity $\mathcal{O}\big(I_{\text{outer}} I_{\text{inner}}(GK^3 + I_P GK)\big)$.
- **Bi-SDR** [7] adopts the semidefinite relaxation (SDR) method to address $\mathcal{P}_2$, with complexity $\mathcal{O}(I_{\text{bis}} I_{\text{IP}} G^3 M^6)$, where I_{bis} and I_{IP} denote the number of bisection and interior-point iterations, respectively.

Unless explicitly noted, the *spatial* curves in the figures refer to Scheme 2.

Fig. 1 illustrates the convergence behavior of Algorithm 1 at a transmit power of $P = 5$ dBW. The proposed method converges within tens of iterations and closely follows the

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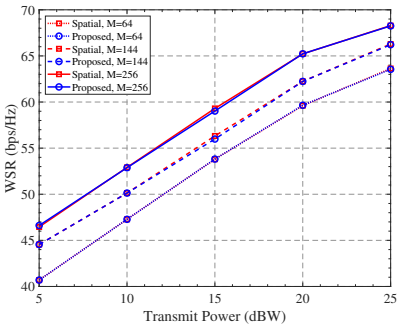


Fig. 2. WSR versus the total transmit power.

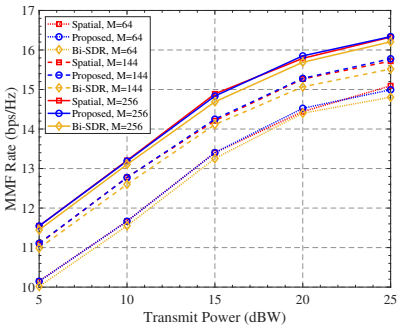


Fig. 3. MMF rate versus the total transmit power.

TABLE II
COMPUTATION TIME (IN SECONDS) OF DIFFERENT APPROACHES UNDER MMF AND WSR CRITERIA.

Criterion	Scheme \ M	8×8	10×10	12×12	16×16
MMF	Bi-SDR	470.32	1855.85	4572.07	19062.14
	Scheme1	144.59	370.77	885.31	3145.22
	Scheme2	24.79	28.64	31.64	133.64
	Scheme3	102.72	100.47	104.55	105.29
	Proposed	19.55	18.85	19.05	19.60
WSR	Scheme1	151.44	336.79	1086.85	3375.12
	Scheme2	24.89	27.75	45.30	124.32
	Scheme3	124.12	130.70	129.70	133.01
	Proposed	20.62	20.70	20.12	20.95

spatial-domain baseline, demonstrating that enforcing $\mathbf{w}_g = \mathbf{G}\mathbf{u}_g$ does not affect the convergence performance.

Fig. 2 shows the ergodic WSR versus the total transmit power for $M \in \{8\times8, 12\times12, 16\times16\}$. Across all array sizes, the proposed scheme achieves the identical performance as the spatial-domain counterpart. Fig. 3 illustrates that the proposed scheme consistently outperforms the Bi-SDR scheme under the MMF criterion, while achieving virtually identical performance to its spatial-domain counterpart.

Table II presents the CPU runtime at a transmit power of 5 dBW. It is observed that the proposed scheme achieves consistently low runtime and shows much weaker dependence on the number of antennas M, whereas the spatial-domain baselines become increasingly expensive as M grows. This highlights the scalability of the proposed scheme and its suitability for massive MIMO LEO communication systems.

V. CONCLUSIONS

In this paper, we proposed a low-complexity precoder design for massive MIMO LEO satellite multicast communications. By exploiting the fact that the optimal precoders lie in a low-dimensional subspace, the high-dimensional spatial-domain problem was equivalently reformulated into a reduced-dimensional form without any loss of optimality. Building upon this reformulation, efficient algorithms with guaranteed convergence were developed under the PDD framework. In particular, a one-time precomputation scheme was introduced to eliminate per-iteration matrix inversions by reusing precomputed factors, thereby further reducing computational complexity. Simulation results showed that the proposed design achieved superior performance with significantly lower complexity, demonstrating its effectiveness for massive MIMO LEO satellite communication systems.

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